SQL Interview Question asked in

Problem Statement

Write a SQL query to find the average transaction amount for each account, but only for those accounts where the average transaction amount exceeds the overall average transaction amount across all accounts.
Also, include the account name in the result set.

Input

account_id	transaction_date	amount
101	2024-01-01	1000.00
101	2024-01-02	1200.00
102	2024-01-03	800.00
102	2024-01-04	1500.00
103	2024-01-05	2000.00
103	2024-01-06	900.00

Output

account_id	Avg_amount
103	1450.000000

Table Creation and Data Insertion script

```
----Table creation Script
CREATE TABLE transactions
 (
   account_id INT,
   transaction_date DATE NOT NULL,
   amount DECIMAL(10, 2) NOT NULL,
  )

--Data Insertion Script
INSERT INTO transactions (account_id, transaction_date, amount)
VALUES
   (101,'2024-01-01', 1000.00),
   (101,'2024-01-02', 1200.00),
   (102,'2024-01-03', 800.00),
   (102,'2024-01-04', 1500.00),
   (103,'2024-01-05', 2000.00),
   (103,'2024-01-06', 900.00)
```

Table Transactions

account_id	transaction_date	amount
101	2024-01-01	1000.00
101	2024-01-02	1200.00
102	2024-01-03	800.00
102	2024-01-04	1500.00
103	2024-01-05	2000.00
103	2024-01-06	900.00

Solution

```sql
select account_id,avg(amount) Avg_amount
from transactions t
group by account_id
having avg(amount)>(select avg(amount) from transactions)
```

Results | Messages

account_id	Avg_amount
103	1450.000000

SQL Script

```sql
select account_id,avg(amount) Avg_amount
from transactions t
group by account_id
having avg(amount)>(select avg(amount) from transactions)
```

Explanation

- First part is to calculate average amount across accounts. Simply using avg() function with grouping by account_id will fetch account wise averge values.
- Second part is to figure out overall average value. We can achieve it using avg() function across table. no group by clause needed here.
- Third part is to compare and find out which accounts are performing better than overall average. We have used having clause against a sub query that returns overall average.